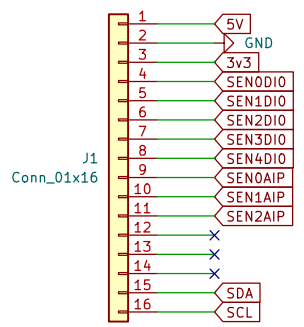


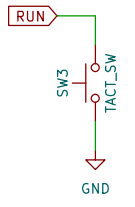
Sensor connector



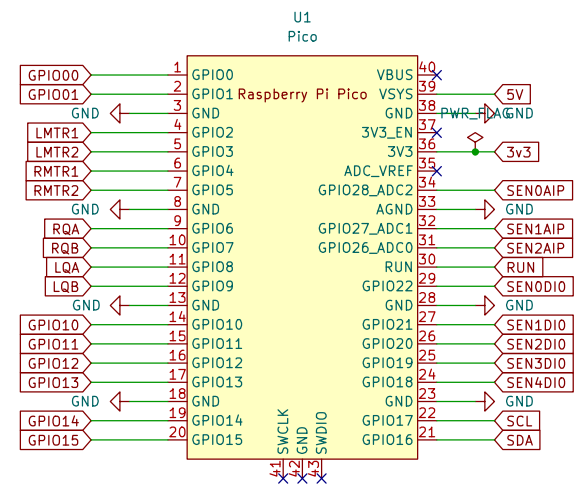
Mezzanine connections

MicroController voltage is 3.3v for any RP2040

PiPico RESET push button

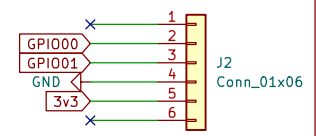


Raspberry Pi Pico / PicoW

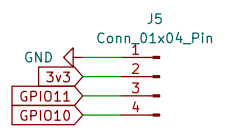


Breakout Connectors

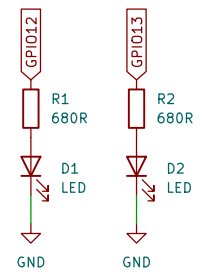
J2 presents GP0 and GP1 with 3v3 and Gnd on pins suitable for bluetooth module to communicate on UART0. The pins can also present I2C0
A vs838 IR receiver could be installed on J2.



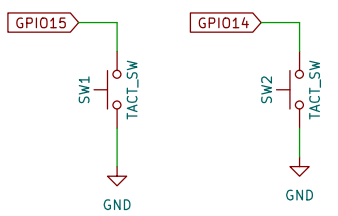
J5 presents GP10 and GP11 for I/O or alternative connection to I2C1 suitable for connecting 0.9" OLED screen or KY-022 IR receiver



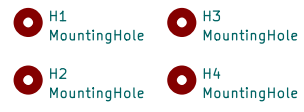
Board LEDs



Board switches/push buttons



MECHANICAL



UKMARS Gemini RP2040 Mezzanine Board Schematic

Sheet: /		
File: UKMARSBOT_RP2040Mezzanine.kicad_sch		
Title: UKMARS Gemini RP2040 Mezzanine V2.0		
Size: A4	Date: 2025-07-07	Rev: 2
KiCad E.D.A. 8.0.8	Id: 1/1	